MICA Commercial

The customer commitment specialist. Explains what a customer ordered, what was promised, what has shipped, and what service or warranty obligation is still open.

Who this is for: product, engineering, operations and anyone who has to explain XELOR to somebody else.
What it covers: the work MICA reasons about, the rules it cannot break, the exact tools it can call, and what is honestly not built yet.
Truth standard: the repository as it stands on 3 August 2026.

MICA

01 · AT A GLANCE

What MICA is for

A role with a fixed tool list, not a general assistant

In one sentence

Explains what a customer ordered, what was promised, what has shipped, and what service or warranty obligation is still open.

2 business areas it speaks for	**2** tools it can actually call	**3** capability families allowed	**0** agents it may delegate to

What reaches it

Customer orders and requested dates · Credit exposure and dispatch state · Ticket, SLA and entitlement state · Supply, plan and quality answers from peers

What it hands back

What is actually committed · Where a promise is at risk, and why · A commercial follow-up work item · A clear line between promised and not yet promised

Capability families it is allowed to touch

sales. · customer. · agent.

Ownership and authority are not the same thing

The areas above are where MICA is the right specialist to ask. The tool table on page 8 is the much smaller set it can invoke at runtime. Owning a business area does not grant runtime power over it — and for ONYX, the supervisor, the gap between the two is the widest in the system.

What sits behind MICA

Records, screens, workflow, controls and hand-offs

Sales	MICA module · sales
Purpose	Turns an accepted customer commitment into a controlled order, dispatch and receivable while preserving the tax and credit decision made at each step.
Core records	Customers, ship-to data, GSTINs, sales orders/lines, credit snapshots and overrides, delivery notes, dispatch quantities, invoices and receivables.
User surface	Orders; Order detail; Customers. The dashboard highlights overdue promises, today-due orders and credit holds from live order data.
End-to-end flow	Create or detect a duplicate customer; draft an order; calculate GST from supplier/place-of-supply state; confirm through the credit gate; reserve nothing automatically; dispatch only available stock; create delivery, invoice and receivable in the same transaction.
Enforced controls	Duplicate detection, immutable credit snapshots, mandatory override reason, server-side tax calculation, dispatch refusal on credit/stock gates, idempotency and atomic stock-plus-invoice posting.
Inputs and outputs	Reads Engineering items and Accounts exposure; asks Inventory to issue stock; hands dated demand to Planning, invoice data to Accounts and fulfilment context to Service.
Current boundary	Order-to-dispatch is live. Leads, opportunities, quotations/tenders, order cancellation, reservations and statutory submission are not complete.
Primary code map	`apps/web/src/modules/sales/manifest.ts` · `apps/api/src/modules/sales/`

How to read these module pages

“Live” means the records, service rules and user/API surface exist in this repository. It does not mean every surrounding enterprise process is complete. The boundary row names what remains illustrative, manual, simulate-driven or outside the MVP.

How MICA's modules connect

A module owns its records; the agent explains the combined evidence

Service

MICA module · csp

Purpose	Keeps customer issues, entitlement, service timing and communication together while separating internal staff access from customer-portal access.
Core records	Customer accounts/users, installed base, warranty/AMC entitlement, tickets, comments, SLA pauses/breaches, complaints, spare requests, knowledge articles, reply drafts and CSAT.
User surface	Tickets; Ticket detail; Spares; Service dashboard; CSAT, plus separate customer-portal APIs.
End-to-end flow	Open and classify a ticket; calculate the response/resolution clock in business hours; check installed-base entitlement; assign and work the case; generate a suggestion if allowed; require staff review before sending; capture complaint, spare and satisfaction outcomes.
Enforced controls	Tenant plus customer-account RLS, business-time recomputation, immutable sent comments, entitlement check, portal/staff permission separation and draft rules that refuse unsupported commitments or leaked internal references.
Inputs and outputs	Uses Sales customer/order context, Inventory spare availability, Quality complaint/NCR context and AI Operations for triage/reply governance.
Current boundary	Ticket, SLA, entitlement, portal isolation and reviewable AI suggestions are live; outbound messaging and automatic warranty population are not integrated.
Primary code map	`apps/web/src/modules/csp/manifest.ts · apps/api/src/modules/csp/`

Cross-module coordination

Sales records what was promised and fulfilled; Service records what happened after delivery. MICA links the same customer, order, installed-base and quality evidence without allowing the portal to cross into another customer's account.

Sales → **Service** → **MICA**bounded evidence → **ONYX / person**decision

How MICA's world actually runs

The business pipeline, not the agent plumbing

The sales pipeline starts at the order. There is no lead, no opportunity and no quotation — by the time XELOR sees a customer, somebody has already decided to buy.

1. **Customer**
   Created against a duplicate check; a suspected duplicate returns 409 with the evidence rather than a silent second record. Carries a credit limit and credit days.

2. **Order (draft)**
   Tax is decided, not chosen: the supplier and place-of-supply GSTINs give the state codes, and one inter-state test drives IGST or CGST+SGST for every line.

3. **Confirm — the credit gate**
   Exposure = receivables outstanding + other confirmed orders. Over the limit does NOT throw; it sets credit_hold. An override needs a written reason, and the database refuses an override without one.

4. **Dispatch**
   Here the hold bites. Four gates plus stock: a short balance aborts the whole dispatch in the same transaction, including the invoice.

5. **Invoice**
   Raised through the Accounts port inside that same transaction. Tax is recomputed for the shipped subset only; Accounts records what Sales calculated and never recalculates it.

6. **Service**
   A ticket runs a business-time SLA clock — Mon–Sat 09:00–18:00 — re-derived from pause intervals, never accumulated into a column.

How much of this MICA can actually see

Of everything above, MICA can read exactly one thing directly — sales orders. The rest of this pipeline is context for judging what it reads; it is not data the agent can reach. Where a mission needs more, it asks the specialist who owns it or it says the evidence is missing.

What the code will not let anyone do

Enforced by constraints and locks, not by wording

The credit gate stops the goods, not the paperwork

Failing the check sets credit_hold and lets the order exist. Dispatch is what refuses — which is where the money actually leaves.

A portal row is fenced twice

Fourteen CSP tables carry a RESTRICTIVE policy on customer_account_id as well as tenant_id, and a staff session writes the setting to empty on every transaction so a pooled connection cannot carry the last request's customer.

A draft cannot promise anything

Reply drafting is refused for commitments, liability admissions, leaked NCR or CAPA numbers, coverage claims with no entitlement check, and any number not already in the thread.

A sent reply is frozen

An AI draft is a comment that is structurally forbidden a sent timestamp. Sending rewrites the author to staff; a trigger then freezes the body, and comments are never deleted.

> **Why this page matters more than the agent pages**
> An agent is only as trustworthy as the system it reads. Every rule here holds whether the request came from a person, a screen, a seeder or MICA — which is precisely why an agent can be given a read tool without also being given a way to do damage.

Where these rules are kept

```
apps/api/src/modules/sales/sales.service.ts
```

```
packages/platform/src/tax/gst.ts
```

```
packages/platform/src/csp/business-time.ts
```

05 · RUNTIME CONTRACT

How MICA takes part in a mission

The engine controls order, parallelism and recovery

Person or signal → **ONYX** scope the mission → **MICA** Commercial → **Capability registry** allow-listed tool → **Domain service** business rules → **Tenant data** + audit

1 · Scope

ONYX turns the goal into a run against a frozen graph version, with a fixed tenant, user, step budget and timeout.

2 · Authorise

Two checks, both required: the tool must be on MICA's allow-list, and the signed-in person must independently hold the permission.

3 · Execute

A registered domain service does the work under the same rules a screen would face. There is no general SQL doorway.

4 · Preserve

Inputs, outputs, events and a checkpoint after every wave, so the run can be explained or resumed.

An agent can narrow authority, never widen it

MICA is a specialist and delegates to nobody. Because the permission check is repeated against the requesting person, a mission can never reach data that person could not have opened themselves.

06 · EXACT CALLABLE SURFACE

Everything MICA can call

This table is the complete list — there is no other door

Capability	Mode	What comes back	Permission required	Needs approval
sales.orders.read	Read	Sales orders	sales.order.read	No
agent.action.dispatch	Execute	Governed work item	agentos.run.operate	Yes

Read · Analyse · Simulate

Return evidence or a reproducible view. Nothing in the business changes.

Draft

Produce reviewable content that is labelled a draft and can never present itself as an approved outcome.

Execute

The only side-effecting mode in the whole system, and the only one that requires an approved human gate above it.

Both locks must open

The agent's allow-list AND the person's permission. Either one failing is a refusal.

What “dispatch” does and does not do

It appends a governed work item naming the responsible specialist, the approval it came from and the exact payload. It does not change a sales order, a stock balance, a production order, a journal or a customer message — a person still does that work.

07 · IN THE FIVE MISSIONS

Where MICA actually appears

Node by node, from the registered graph definitions

Mission graph	What MICA does in it
Cross-functional readiness	Reads sales orders, then assesses commitments
Seven-agent operating review	Reads orders, then the commercial assessment
Controlled action mission	Reads, recommends, and dispatches the commitment-recovery item
Working Capital Review	Reads customer commitments feeding the cash view
QMS & Audit Readiness	Not involved

Reading this honestly

“Not involved” is not a limitation, it is the design. A mission asks only the specialists whose evidence it needs, and a specialist cannot add itself to a graph at runtime. The graph is written down, versioned and content-hashed before the run starts.

The shape every mission shares

Goal → **Parallel reads** → **Assessments** → **Join** → **HEXA verify** → **Human gate** → **Outcome**

08 · WORKED EXAMPLE

The Northstar order, and a control that fires

Real records from the running demo, not an illustration

SO-2627-00004 — 120 PX-400 at ₹52,500, customer reference NPS/PO/10482. Gujarat, so IGST rather than CGST+SGST: the system worked that out from the two GSTINs and nobody chose it.

₹74.34 lakh against a ₹45 lakh limit put the order on credit_hold. It is now override, and the reason is ON the order — eleven of eleven invoices paid within terms, an MD-approved temporary limit, a board note reference.

Later the customer reports a seal weep on TKT-2627-00015. The AI proposed a category and a person accepted it; the AI drafted a reply and a person edited and sent it. The proposal and the acceptance are two separate recorded acts.

The sentence to say out loud

"MICA contributes evidence from its own area, with the source records behind it and its limits stated. It does not claim an action is done until the system has recorded and verified it."

Fair questions to ask while watching

- Which records is this answer standing on?
- Which permission was checked, and against whom?
- Is this a recorded fact, a simulation, a draft, or a completed result?
- Would this step have waited for a person?
- What happens if the service restarts halfway through?

Live today, and deliberately not

The second column is the one worth reading

Working now

- GST place-of-supply and the tax split
- The credit gate with snapshotted limit and exposure
- Dispatch, delivery notes and the invoice
- Business-time SLA, warranty and AMC entitlement
- Triage and reply drafting as suggestions only

Not built — stated plainly

- No lead, opportunity, quotation or tender — the largest commercial gap
- No order cancellation path, though the status exists
- No quality gate on dispatch, and no e-way bill or e-invoice submission from Sales
- Nothing writes warranty rows — they exist only from seed data

Identity boundary	Verified token → tenant and actor
Authority boundary	Agent allow-list AND the person's permission
Action boundary	Side effects need an approved ancestor node
Data boundary	Domain service over a tenant-fenced database
Evidence boundary	Append-only runs, events, checkpoints and audit

Gaps are listed because a guide that hides them fails the first hour of technical due diligence. Every item on the right is either a roadmap decision or a known defect, and both are recorded in the project's own capability-gap register.

MICA on one page

For explaining the agent to somebody new

Its job

Explains what a customer ordered, what was promised, what has shipped, and what service or warranty obligation is still open.

Speaks for

Sales, Service portal (CSP)

Takes in

Customer orders and requested dates · Credit exposure and dispatch state · Ticket, SLA and entitlement state · Supply, plan and quality answers from peers

Gives back

What is actually committed · Where a promise is at risk, and why · A commercial follow-up work item · A clear line between promised and not yet promised

Can call

sales.orders.read, agent.action.dispatch

Cannot do

No lead, opportunity, quotation or tender — the largest commercial gap · No order cancellation path, though the status exists · No quality gate on dispatch, and no e-way bill or e-invoice submission from Sales

Words used on these pages

Agent	A named specialist with a fixed, allow-listed set of tools — not a process that can roam.
Capability	One registered operation with a declared mode, owner and required permission.
Mission graph	A versioned recipe: which steps run, which run together, where it waits for a person.
Checkpoint	A stored snapshot after each wave, used to explain a run and to resume it safely.
Governed work item	An approved, append-only assignment for a human team — not the business change itself.